

① 2.5 - 24

$$U = \left[\begin{array}{ccc|ccc} 1 & a & b & 1 & 0 & 0 \\ 0 & 1 & c & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

↪

$$\left[\begin{array}{ccc|ccc} 1 & 0 & b-ac & 1 & -a & 0 \\ 0 & 1 & c & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

↪

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -a & ac-b \\ 0 & 1 & 0 & 0 & 1 & -c \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

~~~~~

② 2.5 - 40

$$A \rightarrow 4 \times 4 = \begin{bmatrix} 1 & -a & 0 & 0 \\ 0 & 1 & -b & 0 \\ 0 & 0 & 1 & -c \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\left[ \begin{array}{cccc|cccc} 1 & -a & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & -b & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -c & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow{c}$$

$$\left[ \begin{array}{cccc|cccc} 1 & -a & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & -b & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & c \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow{b}$$

$$\left[ \begin{array}{cccc|cccc} 1 & -a & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & b & bc \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & c \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow{a} \begin{bmatrix} 1 & a & ab & abc \\ 0 & 1 & b & bc \\ 0 & 0 & 1 & c \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

③ 2.6 - 13

$$A = \begin{bmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{bmatrix} \rightarrow \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & b-a & c-a & c-a \\ 0 & b-a & c-a & d-a \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & 0 & c-b & d-b \end{bmatrix}$$

$$\hookrightarrow \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & 0 & 0 & d-c \end{bmatrix} = U$$

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \quad \begin{array}{l} a \neq 0 \\ b \neq a \\ c \neq b \\ d \neq c \end{array}$$

④ 2.6-18

$$A = L D U$$

$$A = L_1 D_1 U_1$$

$$L D U = L_1 D_1 U_1$$

$$L_1^{-1} L D U = D_1 U_1$$

lower triax

$$\underbrace{(L_1^{-1} L)}_{\text{lower triax}} \underbrace{D}_{\text{diagonal}} = \underbrace{D_1}_{\text{diagonal}} \underbrace{(U_1 U_1^{-1})}_{\text{upper triangular}}$$

upper triangular  
but they have the diagonal 1  
 $U_1 U_1^{-1} = I$

↳ have to be same.

$$\begin{bmatrix} 1 & a & a \\ 0 & 1 & a \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & b & b \\ 0 & 1 & b \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & & \\ 0 & 1 & \\ 0 & 0 & 1 \end{bmatrix}$$

$$\textcircled{8} \quad \underline{2 \cdot 6 - 28}$$

$$K = \begin{bmatrix} 2 & -1 & & & & \\ -1 & 2 & & & & \\ & & 1 & & & \\ & & & \ddots & & \\ & & & & 0 & \\ & & & & & -1 \\ & & & & & & -1 & 2 \end{bmatrix}$$

$$2^{-2/3}$$

$$= \textcircled{4/9}$$

$$\Rightarrow \begin{bmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & -1 & 2^{3/4} & -1 \rightarrow 0 \\ 0 & 0 & 0 & 0 & -1 & 2 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & 3/2 & -1 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

$$L_1 \begin{bmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & 3/2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 4/3 & -1 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

$$2^{-3/4}$$

$$5/4$$

$$L_2 \begin{bmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & 3/2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 4/3 & -1 & 0 & 0 \\ 0 & 0 & 0 & 5/4 & -1 & 0 \\ 0 & 0 & 0 & -1 & 2 & -p \end{bmatrix}$$

$$\begin{bmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & 3/2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 4/3 & -1 & 0 & 0 \\ 0 & 0 & 0 & 5/4 & -1 & 0 \\ 0 & 0 & 0 & -1 & 2 & -p \end{bmatrix}$$

$$2 - 6/5$$

$$\hookrightarrow \begin{bmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & 3/2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 4/3 & -1 & 0 & 0 \\ 0 & 0 & 0 & 5/4 & -1 & 0 \\ 0 & 0 & 0 & 0 & 6/5 & -1 \\ 0 & 0 & 0 & 0 & -1 & 2 \end{bmatrix}$$

$$\hookrightarrow \begin{bmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & 3/2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 4/3 & -1 & 0 & 0 \\ 0 & 0 & 0 & 5/4 & -1 & 0 \\ 0 & 0 & 0 & 0 & 6/5 & -1 \\ 0 & 0 & 0 & 0 & 0 & 7/6 \end{bmatrix} = U$$

# The row ops not possible to find  $L$  directly !!!

do column ops to find  $L$  directly.

$L =$   $\left[ \begin{array}{c} \phantom{0} \\ \phantom{0} \\ \phantom{0} \\ \phantom{0} \\ \phantom{0} \\ \phantom{0} \\ \phantom{0} \\ \phantom{0} \\ \phantom{0} \\ \phantom{0} \end{array} \right]$

# Also just use octave / matlab.

⑥ 2.7 - 13

a)  $P^3 = I$

$P \neq I$

$1 \rightarrow 2$

$2 \rightarrow 3$

$3 \rightarrow 1$

}

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$P^3 = I$

b)  $\hat{P}^4 \neq I$

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

⑦ 2.7 - 36

$AB = A^{-1}$  if  $A \& B$

① Lower triangular with 1's Yes

② Symmetric matrices X

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 8 & 3+4 \\ 5 & 3+4 \end{bmatrix} = \begin{bmatrix} 8 & 7 \\ 8 & 7 \end{bmatrix}$$

③ positive matrices X

$$\begin{matrix} A \\ \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \end{matrix} \rightarrow \begin{matrix} A^{-1} \\ \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \end{matrix}$$

④ Diagonal invertible Yes

⑤ Permutation Yes

⑥

$$\boxed{Q^T = Q^{-1}}$$

$$A^T = A^{-1}$$

$$B^T = B^{-1}$$

$A^{-1}$  is in the group

$$\boxed{(A^{-1})^T} = (A^T)^{-1} = (\underline{A})^{-1} = \boxed{A}$$

$$(A B)^T = B^T A^T = B^{-1} A^{-1} = (A B)^{-1}$$

Yes

(8) 2.7 - 40

$$Q^T = Q^{-1}$$

a)  $Q^T Q = I$

$$Q = \begin{bmatrix} | & | & & | \\ q_1 & q_2 & \dots & q_n \\ | & | & & | \end{bmatrix}$$

$$\begin{bmatrix} -a_1 & - \\ \vdots & \\ -q_n & \end{bmatrix} \begin{bmatrix} | & & | \\ q_i & \dots & q_n \\ | & & | \end{bmatrix} = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix}$$

$$a_{ii} = \frac{q_{i1}^2 + q_{i2}^2 + \dots + q_{in}^2}{1} = 1$$

$$\|q_i\|^2 = 1$$

b) Similarly, we can show

$$q_i^T q_j \quad \forall i \neq j = \underline{\underline{0}}$$

$$c) \quad q_{11} = \cos \theta$$

$$\begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}$$

$$\begin{bmatrix} \cos \theta & \sin \theta \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = 0$$

$$a \cos \theta + b \sin \theta = 0$$

$$a = -\sin \theta$$

$$b = \cos \theta$$

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \left| \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix} \right.$$

$$\begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \quad \begin{bmatrix} \cos \theta & -\sin \theta \\ -\sin \theta & -\cos \theta \end{bmatrix}$$

Q 3.1-18

a)  $aA + bB \checkmark \quad 0 \checkmark$

$(aA^T + bB^T)$

$[aA_{ij}^T + bB_{ij}^T = aA_{ji} + bB_{ji}] \checkmark \quad \text{True}$

b)  $aA + bB$

$-(aA^T + bB^T) \checkmark \quad \text{True}$

c)  $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

False

⑩ 3.1 - 2.3

• unless already a linear combination

#  $Ax = b$

For  $Ax = b$  to be solvable,  
a linear combination of columns of  
 $A$  or  $C(A)$  must contain  $b$ .

if it doesn't contain  $b$ , then  
 $Ax = b$  is unsolvable and  $C(A)$   
for  $[A \ b]$  will get bigger.

(11) 3.1-36

$$S, T \in V$$

$S+T$  contains all  $(s+t)$

$$a) \lambda_1(s_1 + t_1) + \lambda_2(s_2 + t_2)$$

$$\underbrace{(\lambda_1 s_1 + \lambda_2 s_2)}_{s_3} + \underbrace{(\lambda_1 t_1 + \lambda_2 t_2)}_{t_3}$$

$$s_3 + t_3 \in S+T$$

b)  $S+T$  is a linear combination of subspaces whereas  $S \cup T$  is union of collection of points in  $S$  &  $T$ .

$$S+T = (\lambda_1 s + \lambda_2 t)$$

(12)  $3 - 1 = 32$

$$A \quad [A \quad AB]$$

$$\text{col}_1(AB) = b_{11} \text{col}_1 A + b_{12} \cdot \text{col}_2 A$$

So, all columns of  $AB$  are linear combinations of columns of  $A$

$$\text{Hence } C(A) = C(AB)$$

$$\# \quad C(A^2) < C(A)$$

$$\begin{aligned} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} &= \begin{bmatrix} a^2 + bc & ab + bd \\ ac + cd & bc + d^2 \end{bmatrix} \\ &= \begin{bmatrix} a^2 + \widehat{bc}, & b(a+d) \\ c(a+d) & \widehat{bc} + d^2 \end{bmatrix} \end{aligned}$$

$$c(a+d) = k(bc+d^2)$$

$$a^2+bc = k(b(a+d))$$

$$\frac{c(a+d)}{a^2+bc} = \frac{bc+d^2}{b(a+d)}$$

$$\cancel{bc}(a+d)^2 = \underbrace{(bc+d^2)}_{bc} \underbrace{(\underline{a^2+bc})}_{bc}$$

$$(a+d)^2 = \left(1 + \frac{d^2}{bc}\right) \left(1 + \frac{a^2}{bc}\right)$$

$$\boxed{\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}}$$

$$A = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$